HW 8.

Q1.

(a) For a given c,

power function $\beta(\theta; c) = P_\theta(Y > c)$

$= P_\theta(\max\{X_1, X_2, \cdots, X_n\} > c)$

$= 1 - P_\theta(\max\{X_1, \cdots, X_n\} \le c)$

$= 1 - F_\theta(c)^n$ by iid

where $F_\theta(c) = P_\theta(X_i \le c)$

$$= \begin{cases} 0 & c < 0 \\ \frac{c}{\theta} & 0 < c \le \theta \\ 1 & c > \theta \end{cases}$$

$= (\frac{c}{\theta} \wedge 1) \vee 0$

for a given c,

$\beta(\theta; c) = 1 - ((\frac{c}{\theta} \wedge 1) \vee 0)^n$

$$= \begin{cases} 0 & 0 < \theta \le c \\ 1 - (\frac{c}{\theta})^n & \theta > c \end{cases}$$

(b) size $= \beta(\frac{1}{2}) = 1 - ((2c \wedge 1) \vee 0)^n = 5\%$

$\Rightarrow c = \sqrt[n]{0.95}/2$

(c) p-val $= P_{\frac{1}{2}}(Y > Y_{obs}) = \beta(\frac{1}{2}; 0.48)$

$= 1 - 0.96^{20} \approx 0.558$

retain the H_0 with size 5%. $c \approx 0.499$

(d). p-val $= P_{\frac{1}{2}}(Y > Y_{obs}) = \beta(\frac{1}{2}; 0.52) = 0$.

reject the H_0 with size 5%.

Q2.

before	after	total
922	997	1919

code "before" as 0, "after" 1. (the other coding is fine but H_1 needs to be changed)

Then, $N_1 \sim Bin(1919, \theta)$

$H_0: \theta = \frac{1}{2}$, $H_1: \theta > \frac{1}{2}$

reject H_0 if $N_1 > c$.

$$\text{p-val} = P_0(N_1 \geq N_{obs}) = \sum_{k=N_{obs}}^{1919} P_0(N_1 = k)$$

$$= \sum_{k=997}^{1919} \binom{1919}{k} (\tfrac{1}{2})^k (\tfrac{1}{2})^{1919-k}$$

$$\approx 0.046$$

R code: `1 - pbinom(996, 1919, 1/2)`

(996 $\leftarrow N_{obs} - 1$; $\frac{1}{2} \leftarrow \theta_0$)

reject H_0 with size 5%

retain H_0 with size 1%.

(b) let $X_i \overset{iid}{\sim} Ber(\theta)$

$$X_i = \begin{cases} 1 & \text{death after holiday for } i \\ 0 & \text{death before holiday for } i. \end{cases}$$

by fundamental thm of MLE,

$$\frac{(\hat{\theta} - \theta)}{\sqrt{\hat{\theta}(1-\hat{\theta})/n}} \text{ is } N(0,1)$$

where $\hat{\theta} = \bar{X}_n$

95% approximate CI for θ is

$$\hat{\theta} \pm z_{0.05} * \sqrt{\hat{\theta}(1-\hat{\theta})/n}.$$

$\hat{\theta} = \frac{997}{1919} \approx 0.5195$, $n = 1919$,

then 95% CI is $0.5195 \pm 1.96 * 0.0057$

$\approx [0.508, 0.531]$

Q3.

(a) $\lambda_n = 2 \log \frac{\mathcal{L}_n(\hat{p})}{\mathcal{L}_n(\hat{p}_0)}$

$$= 2 \log \frac{\hat{p}^X (1-\hat{p})^{n-X}}{p_0^X (1-p_0)^{n-X}} \quad \text{where } \hat{p} = \frac{X}{n}$$

$$= 2\left[X \log\left(\frac{X}{n p_0}\right) + (n-X) \log\left(\frac{n-X}{n(1-p_0)}\right)\right]$$

Wilks' thm implies

$\lambda_n \xrightarrow{d} \chi^2_1$ under H_0.

reject H_0, if $\lambda_n > \chi^2_1(\alpha)$ with size α.

↑ α-upper quantile of χ^2_1

(b) under H_0, $\dfrac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}} \dot{\sim} N(0,1)$

then $n(\hat{p}-p_0)^2/(p_0(1-p_0)) \dot{\sim} \chi^2_1$

The Wald test reject H_0 if

$$n\left(\frac{X}{n}-p_0\right)^2/(p_0(1-p_0)) > \chi^2_1(\alpha).$$

Alternatively, reject H_0 if

$$\left|\frac{\frac{X}{n}-p_0}{\sqrt{p_0(1-p_0)/n}}\right| > z_{\alpha/2}.$$

Both have size α asymptotically.

Q4.

$$\lambda_n = 2\log\frac{\mathcal{L}_n(\hat{\beta})}{\mathcal{L}_n(\hat{\beta}_0)}$$

$$= 2\log\frac{\mathcal{L}_n(\hat{\beta})}{\mathcal{L}_n(0)}$$

$$= -\frac{1}{\sigma^2}\left(\sum_{i=1}^n (y_i - x_i\hat{\beta})^2 - \sum_{i=1}^n y_i^2\right) \qquad \cdot\ \hat{\beta} = \frac{\sum x_i y_i}{\sum x_i^2}$$

$$= -\frac{1}{4}\left[(\Sigma x_i^2)\hat{\beta}^2 - 2(\Sigma x_i y_i)\hat{\beta}\right]$$

$$= \frac{1}{4}\left(\sum_{i=1}^n x_i y_i\right)^2 / \sum_{i=1}^n x_i^2$$

reject H_0 if $\lambda_n > \chi^2_1(\alpha)$.